

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 2

Mock Interview

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO5: Students will be able to evaluate advanced machine learning concepts such as reinforcement learning, Monte Carlo methods, sampling techniques, policy iteration, value iteration, and temporal difference learning. They will understand computational learning theory concepts including VC dimension, sample complexity, and mistake-bound analysis. Students will be capable of selecting and optimizing advanced learning strategies for intelligent systems and complex applications. (BL-6)

Assessment Tool Weightage: 20%

QUESTIONS:

Q1. Autoencoder Design

- What is an autoencoder and what are its main components in a neural network architecture?
- How would you design an autoencoder for reducing the dimensionality of a high-dimensional dataset?
- What factors would you consider when deciding the size of the encoder and decoder layers?
- Why is the bottleneck layer important in an autoencoder design?
- How would you choose an appropriate activation function for the encoder and decoder of an autoencoder?

Q2. Stochastic Encoders and Decoders

- What is a stochastic encoder, and how does it differ from a deterministic encoder?
- Why is randomness introduced into the encoding process of a stochastic autoencoder?
- How does a stochastic decoder reconstruct data from a probabilistic latent representation?
- What are the advantages of using stochastic encoders and decoders in generative models?
- How does a stochastic encoder represent uncertainty in the input data?

Q3. Deep Belief Network vs Deep Boltzmann Machine

- What is the main architectural difference between a Deep Belief Network (DBN) and a Deep Boltzmann Machine (DBM)?
- How are connections between hidden layers organized differently in a DBN and a DBM?
- Why is a DBN considered a hybrid model containing both directed and undirected connections?
- How does a DBM differ from a DBN in terms of the direction of connections between layers?
- How does the training procedure of a DBN differ from that of a DBM?

Q4. Generative Adversarial Networks

- What is a Generative Adversarial Network, and what are its two main components?
- How does the generator in a GAN create new data samples?
- What is the primary role of the discriminator in a GAN?

- How do the generator and discriminator compete with each other during GAN training?
- Why is balancing the generator and discriminator important for stable GAN training?

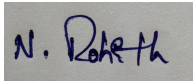
Q5. Integrated Deep Learning Solution

- What do you understand by an integrated deep learning solution, and what are its major components?
- How would you combine different deep learning models to solve a complex real-world problem?
- How would you decide whether to use an autoencoder, DBN, DBM, or GAN in an integrated solution?
- What factors should be considered when designing an end-to-end deep learning pipeline?
- How can feature extraction and representation learning be integrated into a deep learning solution?

Code of Conduct:

I **NALIPI REDDY ROHITH/192425115** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Mock Interview

Q1. Autoencoder Design

1. What is an autoencoder and what are its main components?

An **autoencoder** is a neural network that learns to represent input data in a compressed form and then reconstruct the original data.

Its main components are:

- **Encoder** – converts input into a smaller latent representation.
- **Bottleneck/Latent layer** – stores the important features.
- **Decoder** – reconstructs the original input.

2. How would you design an autoencoder for dimensionality reduction?

I would use a gradually decreasing encoder structure, for example:

100 → 64 → 32 → 10

The 10-neuron layer becomes the compressed representation. The decoder reverses this process:

10 → 32 → 64 → 100

The model is trained by minimizing reconstruction error, usually **MSE**.

3. What factors decide the size of encoder and decoder layers?

I would consider:

- Number of input features.
- Complexity of the dataset.
- Amount of information that must be retained.
- Computational resources.
- Required compression level.

Too few neurons may lose important information, while too many may reduce the benefit of dimensionality reduction.

4. Why is the bottleneck layer important?

The bottleneck forces the network to learn only the **most important features** of the input. It provides a compact representation that can be used for dimensionality reduction, visualization or further machine-learning tasks.

5. How would you choose activation functions?

For hidden layers, **ReLU** is commonly used because it provides non-linearity and trains efficiently.

For the decoder:

- **Linear** can be used for continuous numerical data.
- **Sigmoid** can be used when outputs are normalized between 0 and 1.
- **Softmax** can be used when reconstructing categorical probability distributions.

Q2. Stochastic Encoders and Decoders

1. What is a stochastic encoder?

A stochastic encoder does not produce one fixed latent vector. Instead, it produces a **probability distribution**, usually described by a mean and variance.

A deterministic encoder gives:

Input → Fixed latent vector

A stochastic encoder gives:

Input → Probability distribution → Sampled latent vector

2. Why introduce randomness?

Randomness helps the model learn a **distribution of possible representations** rather than memorizing one representation.

This is especially useful in generative models because it allows the model to generate different samples.

3. How does a stochastic decoder reconstruct data?

The decoder receives a sampled latent vector and converts it into a reconstructed output.

Input



Encoder



Mean + Variance



Sample latent vector



Decoder



Reconstructed Data

4. Advantages in generative models

Stochastic encoders and decoders:

- Represent uncertainty.
- Generate diverse outputs.
- Learn probability distributions.
- Improve generative capability.
- Help create smooth latent spaces.

5. How does a stochastic encoder represent uncertainty?

It represents each input using parameters such as **mean (μ)** and **variance (σ^2)** instead of a single fixed value.

For example:

$$z \sim N(\mu, \sigma^2)$$

A larger variance indicates greater uncertainty in that part of the representation.

Q3. Deep Belief Network vs Deep Boltzmann Machine

1. Main architectural difference?

A **DBN (Deep Belief Network)** consists of stacked RBMs, where the connections between higher layers are generally directed.

A **DBM (Deep Boltzmann Machine)** has undirected connections between neighboring layers.

2. How are hidden-layer connections organized?

In a DBN:

Hidden 3

↑

Hidden 2

↑

Hidden 1

↑

Visible

In a DBM:

Hidden 2 ↔ Hidden 3

↕

Hidden 1

↕

Visible

The DBM uses **bidirectional/undirected connections**.

3. Why is DBN a hybrid model?

A DBN combines different types of connections. The lower portion can be represented using undirected RBM connections, while higher-level relationships are represented using directed connections.

Therefore, it combines **undirected and directed probabilistic structures**.

4. How does DBM differ in connection direction?

A DBM is an undirected graphical model. Connections between adjacent layers work in both directions.

There is no fixed direction such as:

Layer 1 $\rightarrow$ Layer 2

Instead:

Layer 1 $\leftrightarrow$ Layer 2

5. How does training differ?

DBN:

- Usually trained layer-by-layer.
- Each RBM is pre-trained separately.
- The complete network can then be fine-tuned.

DBM:

- Requires joint learning of multiple hidden layers.
- Training is generally more computationally expensive.
- Approximation methods are often required.

Q4. Generative Adversarial Networks

1. What is a GAN?

A **Generative Adversarial Network (GAN)** is a generative deep-learning model consisting of two networks:

- **Generator**
- **Discriminator**

They compete with each other during training.

2. How does the Generator create new samples?

The Generator receives a random noise vector:

[
 $z \rightarrow G(z)$
]

and transforms it into a synthetic sample.

For example:

Random Noise

↓

Generator

↓

Synthetic Image

The generator learns to create samples that resemble the real training data.

3. Role of the Discriminator?

The Discriminator determines whether an input sample is:

- **Real** – from the training dataset.
- **Fake** – generated by the Generator.

It acts like a binary classifier.

4. How do they compete?

The Generator tries to **fool the Discriminator** into believing fake samples are real.

The Discriminator tries to correctly identify fake samples.

Generator → Fake Image → Discriminator

↓

Real / Fake

Through repeated competition, the Generator gradually improves.

5. Why is balancing important?

If the Discriminator becomes too strong, the Generator receives very little useful learning signal.

If the Generator becomes too strong too quickly, the Discriminator cannot learn effectively.

Therefore, both networks should improve at a suitable rate. Proper balancing helps achieve **stable GAN training** and produces better-quality samples.

Q5. Integrated Deep Learning Solution

1. What is an integrated deep learning solution?

An integrated deep learning solution combines **multiple models or techniques** into one pipeline to solve a complex real-world problem.

Major components can include:

- Data preprocessing.
- Feature extraction.
- Representation learning.
- Prediction/generation.
- Evaluation.

2. How would you combine different models?

Different models can perform different tasks.

For example:

Raw Data



Autoencoder



Feature Representation



CNN / DBN



Feature Learning



GAN



Synthetic Data



Final Classifier

Each model performs the task for which it is most suitable.

3. How do you choose Autoencoder, DBN, DBM or GAN?

Model	Suitable Use
Autoencoder	Dimensionality reduction and feature learning
DBN	Hierarchical representation learning
DBM	Complex probabilistic representation
GAN	Generating realistic synthetic data

The choice depends on the application, dataset size, computational resources and desired output.

4. Factors in an end-to-end pipeline

Important factors include:

- Dataset quality and size.
- Preprocessing requirements.
- Model complexity.
- Computational resources.
- Training time.
- Overfitting.
- Accuracy and other evaluation metrics.
- Deployment requirements.
- Interpretability.

5. How can feature extraction and representation learning be integrated?

A feature-learning model such as an **autoencoder** can first convert raw high-dimensional data into a compact representation.

Raw Data



Encoder



Latent Features



Deep Learning Model



Prediction

For example, an autoencoder can extract compact features from sensor data, and those features can then be given to a classifier for defect detection.

This reduces unnecessary information and allows the final model to focus on the most useful features.